

4. EMERGENCY SHUT-DOWN PROCEDURE

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4. EMERGENCY SHUT-DOWN PROCEDURE

4.1 Summary of Emergency Shutdown Operations

4.1.1 Summary

In cases where it has become impossible suddenly to continue safe and stable operation of the PTA unit due to occurrence of an unforeseeable accident or where it is judged to be impossible to continue said operation, the flows of raw materials, auxiliary raw materials and utilities are stopped systematically and all facilities are stopped safely and promptly.

Judgment of the state of the accident, instructions on urgent operation and confirmation of such operation are in the hands of the foreman of the shift concerned.

When confronted with the standpoint to give instructions on emergency operation or to operate, all personnel, especially the foreman, should grasp the state of the accident accurately and conduct appropriate operation skillfully with priority given to safety first, environmental security and prevention of the occurrence of a secondary accident.

Emergency operation is classified into two in terms of sequence, 1st operation and 2nd operation.

The 1st operation is the most urgent operation with utmost importance given to the security of safety of the PTA unit. For that reason, the interlock circuit of "PTA Unit All Shut Down" is provided in order to perform the all urgent operations automatically and simultaneously.

Consequently, PTA unit can be stopped safely and promptly by operation of the interlock circuit of "PTA Unit All shutdown". This interlock switch is located on the main control panel and basically, when electric power failure and instrument air failure is occurred, this interlock

switch is operated by board man.

However, besides, in case where the emergency situation is occurred in PTA unit such as a large amount leakage or fire, etc., this interlock switch is operated also.

The 2nd operation is the measure aimed mainly at safety and it is conducted after completing the 1st operation.

In order to become able to perform this urgent operation satisfactorily, all personnel are required to master measures and operations for coping with emergency situations by accumulating practical training as well as desk training on a day-to-day basis, assuming a variety of accidents.

A list of urgent operations is put up in the control room for the purpose of all personnel's confirmation of urgent operations from time to time.

The restart-up of the PTA Unit after an emergency shutdown must be performed under the direction of the foreman after performing flushing adequately and making sure of safety.

4.1.2 Electric Power Failure

1) Electric power system of the PTA plant consists of the 2 feeders of 6.6 KV-TA and PTA and 6.6KV-TC-201.

6.6 KV-TA & PTA feeder for the rotary equipment of Purchased Electric Power and 6.6 KV-TC-201 feeder for the process air compressor of TA unit is the exclusive supply of TC-201. In cases where 6.6 KV-TA & PTA feeder fails, all of rotary equipment in the PTA unit stop.

In cases where instantaneous power failure, the rotary equipment having *TDR are not affected and not stopped. Therefore immediately restart the stopping rotary equipment (not haveing TDR) and continue the normal operation.

*TDR Time Delay Relay;

To be automatically reaccelerated on restoration of power source after instantaneous power failure.

2) The rotary equipment of 6.6 KV feeder is as follows; (The equipment having TDR is shown with * mark.)

a) 6.6 KV-TA & PTA (Purchased Electric Power)

*PP-101A, B	PP-102A, B, C	PP-102-OP(A,B,C)
PP-103A, B	*PP-201A,B	PP-304A, B
PP-402A, B	*PP-501A, B	*PP-502A, B
*PP-503A, B	*PP-504A, B	PP-601A, B, C
*PJ-101	*PJ-301	*PJ-302
*PJ-303	*PJ-304	*PJ-401
*PJ-402	*PJ-404	*PJ-404-1
PX-401A, B, C	*PC-201A, B	PC-601
PC-602	*PJ-901	
*PP-404A, B	PP-405A, B	*PP-406A, B
PP-410	PP-600	PP-603
*PP-901	PM-101-1	PM-101-2
PM-102	PM-105	PM-401A, B, C
PM-402A, B	PM-402A-1/B-1	PM-404-1
PM-404-2	PM-405	*PM-406

PM-901	PM-902	PZ-100
PZ-403	PC-402	*PC-403A
*PC-901	*PC-901-OP	PC-901PM
PZ-800A, B, C	PZ-903	PP-902A, B
TP-802A, B		
Light	Instrumentation	

4.1.2.1 Electric Power Failure of 6.6 KV Feeder

In this case, the whole PTA plant is shut down, therefore, the interlock switch of "PTA Unit All Shut Down" shall be operated immediately by the board man.

Remarks: a) After the operators in the plant confirmed the electric power failure, all of the operators gather in the CCR to get instruction.

b) The rotating equipment of TA and PTA units are connected to the same feeder of electricity.

1) Light

Electric power is backed up from battery to emergency lights.

2) Instrumentation

Electric power is backed up from battery to field instruments, interlock system and DCS in the control room for approx. 0.5 hr.

Electric power to hand set speaker is backed up from battery.

3) Feed and Reactor

- CTA feed to PD-101 stops, and the flow of recycle solvent from PD-501 stops also.
- The agitator PJ-101 stops.
- The Feed and Reactor system is left to stand with the pressure maintained and the liquid retained.

1st operations

- Operations are performed by operating the interlock switch of "PTA Unit All Shut Down" for the purpose of maintaining the pressure of CTA slurry and retaining it inside.

2nd operations

- Close the discharge valve of each pump and lock the local switches.
- Close the block valves of the control valves.

- Close the inlet valves of each user of purge water.

4)Crystallizer

- All the agitators of the Crystallizers stop.
- The flow of 10BW to the PE-302 heat recovery systems stops.
- The flow of purge water (HPW, MPW and LPW) to the level gauge, LV, and PV of each Crystallizer stops.
- The feed from PD-201 stops, the PV and LV of each Crystallizer close, and PP-304A(B) stops. This is placed the Crystallizers to the state in which the pressure is maintained in them.

1st operations

- Operations are performed by actuating the interlock switch of "PTA Unit All Shut Down" for the purpose of maintaining the pressure of CTA slurry and retaining it inside.

2nd operations

- Close the discharge valve of each pump and lock the local switches.
- Close the block valves of LV-2312 of the PE-302.
- Close the inlet valve of each user of purge water.

5)Separation & Drying

- The rotary equipment stops and the interlocks of the Silo, Dryer, RVF and Centrifuge systems are actuated.
- The agitators PJ-401, PJ-402 and PJ-404 stop.
- The interlock of PT-401 is actuated, FV-2408 closes.
- The waste water system downstream of PD-403 gets into the state in which the pressure is maintained and the liquid is retained.
- The Dryer is left with steam flowing in it.

1st operations

- Operations are performed by operating the interlock switch of "PTA Unit All Shut Down".

2nd operations

- Close the discharge valve of each pump and lock the local switch.
- Close the block valves of TV-2401, PV-2401A & B.
- Close the inlet valve of each user of purge water.

6) Solvent and sealing and flushing water

- In the PT-501 system, the blowing of 3S to the PZ-502 vent starts due to the actuation of the interlock of PIS-2510.
- PD-501 gets into the state in which the liquid is retained.
- The receiving of DIW stops.
- The flow of all of purge water DIW, LPW, MPW and HPW stops.

1st operations

- Operations are performed by operating the interlock switch of "PTA Unit All Shut Down".

2nd operations

- Close the discharge valve of each pump and lock the local switches.
- Close the inlet valve of each user of purge water.

7) Hot oil

- As the result of the stoppage of PP-601A, B(C), hot oil circulation stops.
- Due to the actuation of the interlock, in the PF-601 system, SDV-2610A, B (Main burner), SDV-2609A, B (PILOT burner) close. Also, SDV-2611 closes.
- As the result of the stoppage of PC-601 and PC-602, HCSV-2616, opens and HCSV-2603 switch to by-pass side of PE-605 (HCSV-2603 opens), and SDV-2613 opens, resulting in the blowing of snuffing steam.
- Due to the actuation of load down, FV-2602 opens to the set degree of opening.
- The flow of 10BW to PD-1205 stops.

1st operations

- Operations are performed by operating the interlock switch of "PTA Unit All Shut Down".

2nd operations

- Close the discharge valve of each pump and lock the local switches.
- Purge the fuel gas line with N, if necessary.

8) Silo

- Bag Filter pulse unit stops.

1st operation

- Secure N for safety use. (CTA unit operator)

9) PTA residue recovery

- The rotary equipment stops.
- The drum is maintain and the liquid is retained.

1st operations

- Operations are performed by actuating the interlock switch of "PTA Unit All Shut Down".

2nd operations

- Close the discharge valve of pump and lock the local switch.
- Close the inlet valve of purging PQ and sealing PQ.

10) In case of total electric power failure , the EDG is started immediately.

(Starting of the EDG is needed about 40 seconds.)

After confirmation of the starting of the EDG, restart the agitators to prevent of sedimentation of TA in each drum.

After conduct the action of 1st operations and 2nd operations, or feed and reactor section and each drum perform to treatment with that works at the same time.

a) Start the EDG.

b) Introduce FLW to the CCW line for agitators.

- Close HZV-3301.
- Open HZV-3302.

Remarks: • Ensure the cooling water of agitators and pumps.

• Confirm that whether FLW is available or not.

c) Start each agitator.

(PJ-101, 301 ~ 304, 401, 402 and PJ-901)

Remarks: After PJ-301 ~ 304 and PJ-401 are ensured the seal water of shaft, start agitators.

If the seal pressure has been decreased, start PP-504A(B), and after ensuring the seal pressure, start agitators.

d) Perform to purge TA slurry from the feed preheaters and the reactor.

• Start PP-502A(B), PP-103A(B) and PP-504A(B).

• After starting PP-102A(B, C), purge the slurry from PE-101A ~ F and PD-201 to PD-301.

Letting the slurry out is completed by means of solvent operation about 15 minutes.

If necessary, start PP-601A(B, C), using the remaining heat in the Hot oil to keep the temperature of feed and reactor system.

• After completion of purging, stop PP-102A(B, C), PP-103A(B) and PP-502A(B).

PP-504 is continued to operate in order to ensure the seal water for the each agitator.

(PJ-301 ~ 304, PJ-401)

• Restart PX-401A, B, C

Remarks: To prevent the leakage from the part of mechanical seal.

11) Operations to be performed after the power supply is restored in a short time.

The PTA Unit shall be started up in sequence of following the start-up procedures. In doing so, attention should be paid to the following points:

- a) Securing utilities
 - b) Establishment of the LPW, MPW, HPW and DIW systems
 - c) In the Drum retaining slurry, there is the possibility that TA has settled down after the stoppage of the agitator. Therefore, the agitation is started. If the agitator can't be started, dilute the slurry by adding MPW and discharge the slurry from the bottom and try to start again.
 - d) Establishment of the hot oil system
 - e) Establishment of the preheater
 - f) Switch the suction of PP-102A, B(C) to LPW and start PP-102A, B(C).
- 12) Following operations are performed if it is expected that the restoration of power supply will be delayed
- a) Close the inlet valve of PD-201 and blow down the holdups in PE-101A to F.
 - b) The holdup in PD-201 shall be drained to PD-301 by opening PV-2201 slightly while keeping the pressure by using H₂ compressor.
 - c) Proceed to shut down all the systems and blow down the holdup.

Remarks: These works are very dangerous when the holdup slurry is discharged to the sewer from the pressure vessel so that it is necessary to pay the closest attention.

4.1.3 Stop of Utility

4.1.3.1 Stop of 25S

Under normal operation, receipt of the 25S is a small quantity. In case of stoppage of 25S PTA unit it is not necessary to shut down immediately.

However, if it continues to stop condition for a long time, each part of the process might be lowered the temperature, which causes plugging of lines and equipment. In such case PTA unit is shifted to Recycle Solvent Operation.

After confirming the operating conditions, it shall be judged whether the PTA unit continues to operate or not.

1) Main users of 25S

- a) Shell side of PE-101C, PE-503 and PD-1206
- b) Line trace (PE-101B ~ F, etc.)
- c) Valve jacket (LV-2303, etc.)

4.1.3.2 Stop of 15S

15S receives from the OSBL.

10S, 3S and 1S steams are generated from 15S by reducing its pressure in the OSBL.

Therefore in the event of the stoppage of 15S, also 10S, 3S and 1S steams are stopped, necessitating the recycle solvent operation.

1)Main users of 15S

a)TE-901, TE-1133 in the TA unit.

2)Main users of 10S

a)PE-101A, B, PE-401 and PE-504

b)PLH-401, PLH-402

c)Line trace (PE-101A ~ PP-102, etc.)

3)Main users of 3S

a)PM-404 tube

b)Shell side of PE-502, PE-401

c)Line trace (PM-401A ~ C etc.)

4.1.3.3 Stop of IA

In the event of the stoppage of IA supply, the control valves are actuated to the safe side. Also, the indications on the instruments using IA become zero.

As conceivable causes of the stop of supply of IA, there are cases of an extended stoppage of the IA compressor due to a power failure, breakage of piping and miss-operation of a valve.

In this case, the interlock switch of "PTA Unit All Shut Down" shall be operated immediately by the board man.

The process conditions just after the IA stoppage are as follows. (Before actuation of All SD switch)

1)Condition of the utility systems

1-1)The flow of N and steam in ISBL stop.

1-2)As all LV's in the Flash Drum system close, the system gets into the state in which the pressure is maintained and the liquid is retained.

2)Condition of the process systems

2-1)Feed and Reactor system

a)As the flow of CTA pneumatic conveying gas stops, the receiving of CTA stops.

b)PM-101-1, -2, PM-102 and PM-105 continue operation, and the CTA feed to PD-101 continues.

c)The flow of recycle solvent except for the use for purging the PM-105 chute stops.

d)The PD-101 system proceeds to circulation operation.

e)The PE-101A ~ F to PD-201 system get into the state in which the pressure is maintained and the liquid is retained.

f)The H₂ to Reactor stops and PC-201 proceeds to circulation operation.

g)The PP-102 enters into shut-off operation.

h)Heat recovery system of shell side of feed section

get into the state which pressure is maintained and the liquid is retained.

2-2)Crystallizer system

- a)All PV's and LV's in the Crystallizer system close, and PD-301 to PD-304 get into the state in which the pressure is maintained and the liquid is retained.
- b)PV-2201 closes, and the feed to PD-301 stops.
- c)The generation of steam in PD-1202 stops.
- d)As each PV for MPW and LPW opens, the pressure of purge water falls, and purging stops.
- e)The supply of heat source of the purge water heater stops.

2-3)Separation and Drying system

- a)Each control valve closes (except LV-2410A)
- b)The feed to the Centrifuges, RVF and the flow of reslurry water stop.
- c)The DIW feed to the N scrubber of the Dryer system stops and LV opens fully. As a result, the scrubbers become empty.
- d)The supply of steam for heating the Dryer stops.
- e)In the PTA conveying system, FV closes, and the rotary valve, Dryer, etc. are entered to a stop due to the actuation of the low flow interlock. Therefore, the Dryer stops with a holdup in it.
- f)PV-2401A and B close. The pressure in the PM-401 system is maintained.
- g)PD-403 and PD-404 get into the state in which the liquid is retained, and LV-2403A closes. (LV-2403B opens.)
- h)The flow of each purge water stops.

2-4)Process water and sealing and flushing water system

- a)Flash vapor of PT-501 to PD-501 stops because LCV-2510B closes. Also, SDV-2510B opens.
- b)As PV for purge water opens, the pressure of purge

water lowers.

- c)The supply of DIW to PD-502 stops because LV closes.
As PV-2502, PV-2503 opens, PP-502 enters into circulation operation.
- d)As the TV of the purge water heater closes and the supply of heating steam stops.
- e)The supply of LPW to PD-501 stops. And as the supply of recycle solvent to PD-101 stops and LV closes, the level of PD-501 becomes high, and at the same time, PP-501 enters into circulation operation.
(establishment of the bottom circulation line)

2-5)Hot oil system

- a)All the burners of PF-601 go out.
However, PC-601 and PC-602 continue operation.
- b)Hot oil circulation continues because PP-601 does not stop. However, as FV-2601 opens, the pressure of the header lowers.
- c)The Hot oil supply to PE-601 stops, and the 10BW supply to PD-1205 also stops.
- d)As the TV of each of PE-101E & F, PLH-201 and PLH-301 closes, the flow of hot oil stops.
- e)As PV-2608A closes and PV-2608B opens, the flow of sealing N to PD-601 stops, and PD-601 gets into the depressured state. Also, as PV-2600 close, the flow of sealing N to PTK-600 stop.

2-6)Silo system

- a)In the case of 7N being used for each pneumatic conveyor line, pneumatic conveying stops because FV-2405 closes.
- b)Due to the actuation of the interlock, all the rotary valves for delivery stop.
- c)All the two-way changeover valves in the PTA pneumatic conveyor line maintain it's position and becomes inoperative.

d)The pulse unit of the Bag Filter (PM-803) of each Silo does not stop.

2-7)Residue recovery system

a)Each control valve closes.

b)The Fresh Q feed to the PD-901 stop. Therefore, the PD-901 remains with holdup in it.

c)The residue transfer to the TD-102A stops. As a result, PP-902A and B become circulation.

To ensure the safe process conditions from above mentioned situation, the interlock switch of "PTA Unit All Shut Down" and "Depower supply switch" shall be turned on immediately.

3)In cases where the supply of IA is restored in a short time

a)Confirm the action of each instrument and control valve.

b)After inspecting each purge line, put the purge water systems into operation in sequence.

c)Ignite the PF-601 burners and increase the temperature of the hot oil system.

d)Heat up PE-101A to F.

e)Switch the suction of PP-102 to LPW and proceed to hot water operation by starting PP-102.

4.1.3.4 Stop of 9N

When the supply of 9N stops, it is possible to introduce 7N by connecting the hose temporarily.

1) Users of 9N

- a) PD-401, PD-403 Make up
- b) PD-501
- c) PE-404 Vent
- d) PD-502
- e) PM-101
- f) O₂ Analysis box for AI-2400A/B
- g) PX-401A, B, C

4.1.3.5 Stop of CCW

When the supply of CCW stops, it becomes impossible to secure cooling water for major equipment of both TA and PTA Units. In the event of the complete stoppage of CCW, the interlock switch of "PTA Unit All Shut Down" shall be operated immediately by the board man.

1) Users of CCW

- a) PP-102A, B & C oil coolers
- b) PC-201A & B oil coolers
- c) PX-401A, B & C oil coolers
- d) PC-901 oil coolers
- e) PP-601A, B & C motor coolers
- f) PP-503A & B oil coolers
- g) PJ-301, PJ-302, PJ-303, PJ-304, PJ-401
- h) PP-101A/B
- i) PP-304A & B, PP-404A & B
- j) PP-502A & B, PP-501A & B, PP-410
- k) TP-1201A & B
- l) TP-1203A & B
- m) TP-1210A & B
- n) PE-405, PE-406
- o) PE-407, PC-402
- p) PLC-101, PLC-201, PLC-402, PLC-801
- q) PJ-101, PJ-402

4.1.3.6 Stop of supply of FLW

When the supply of FLW stops, it is possible to introduce DIW by connecting the hose temporarily.

1)Users of FLW

a)Utility stations in the PTA Plant

b)Purge water for each mechanical seal of PP-901,TP-800A/B and TP-802A/B.

c)PZ-404, PZ-501, PZ-600, PZ-604 and PZ-905

d)AI-1800,AI-1802, AI-1803, and AI-1806.

e)TZ-1222 and TZ-1224

f)PD-404 top spray and purge water for the grand of the bottom of PJ-404.

g)PM-101 vent

4.1.3.7 Stop of DIW

When the supply of DIW stops, the level of PD-501 falls to zero in a short period of time, and the supply of the solvent indispensable to the PTA Unit process stops.

In this case, CTA feed is immediately stopped, and operation is quickly switched to recycle solvent operation to purge CTA from the Feed and Reactor system.

While taking care to save the consumption of flushing water, the recycle solvent operation is continued as long as possible by utilizing the solvent in PD-501 and PD-502 in order to flush the system. In this situation bottoms of PT-501 shall be recovered into PD-501 in order to make up solvent. Subsequently, the Feed to Crystallizer systems are brought into the state in which the pressure is maintained and the liquid is retained by operate the interlock switch of "PTA Unit All Shut Down".

1) Users of DIW

- a) PT-401 (FV-2408)
- b) PD-502 (LV-2502)
- c) Purge for mechanical seal of PC-403A and PD-405's stand line.
- d) For flushing the PJ-301, PJ-302, PJ-303, PJ-304, PJ-401 and TJ-201(TA unit) mechanical seal (from PP-504)
- e) Discharge line of PP-402. (FV-2412)
- f) Make-up for TD-1201

4.1.3.8 Stop of H₂

PD-201 is the process unit that carries out chemical reaction (hydrogenation Reaction) in PTA unit.

CTA transported from TA unit is mixed with solvent, converted into TA slurry and fed to PD-201 with its flow rate regulated by FIC-2201A and B(C).

On the other hand, H₂ received from OSBL is fed to PD-201 with its flow rate adjusted by FICQK-2202.

As the result, impurities (ex. 4CBA) in CTA are hydrogenated in the Reactor and converted into water-soluble p-TA.

So, in case of stoppage of H₂, PTA product leads to produce off-spec product. And also, in case that H₂ partial pressure is very low, decarbonylation reaction may take place and care has to be taken fully on this point.

1) Operations to be performed

a) Switch over to recycle solvent operation.

b) Switch H₂ to N at the suction of PD-202.

c) Switch the destination of the discharge of PM-404 from PTK-800A, B or C to TTK-401.

4.1.3.9 HO Failure (PF-601)

HO is mainly used as a heating source in PE-101E & F to convert TA slurry into TA solution.

For that reason, when the furnace PF-601 stops, feed preheater PE-101E & F becomes impossible to get heat from H.O. system.

As the result, dissolution of TA slurry becomes inadequate and TA begins to deposit, causing the plugging of the lines and equipment.

In this case, CTA should be immediately stopped, and the operation is quickly switched to recycle solvent operation, continuing at least 15 minutes to purge TA slurry from the preheaters and the reactor.

1) Uses of H.O.

- a) PE-101E & F
- b) PLH-201
- c) PLH-301
- d) PE-601
- e) PD-301 Discharge line jacket
- f) PD-302 Discharge line jacket
- g) PV-2201 jacket
- h) PD-201 Jacket of bottom valve
- i) PE-101E ~ PD-201 line trace
- j) PD-201 ~ PD-301 line trace

4.2 Emergency Shutdown Operations

4.2.1 Electric Power Failure

4.2.1.1 Electric Power Failure of all feeders of 6.6 KV-TA & PTA and TC-201

4.2.1.1.1 Summary

When the power of all 6.6 KV Feeder fails, all rotary equipment in the PTA is stopped, necessitating an emergency shutdown.

4.2.1.1.2 Procedure

1) 1st operations

Bring each equipment into the state in which the pressure is maintained and the liquid is retained.

1-1) Operate the interlock switch of "PTA Unit All Shut Down".

Remarks: • Do not operate "Depower Supply switch".

• Refer to the interlock circuit for the detailed working conditions.

1-2) Close the block valve of FV-2202 in the H₂ feed line.

1-3) Close FV-2602 and FV-2601 on manual.

1-4) Switch off the rotary equipment, having "MZ" on DCS.

Remarks: When electric power source is restored, each rotary equipment is started automatically so that turn off the switch on DCS.

2) 2nd operations

2-1) Feed and Reactor

a) Turn off the heating and pressuring sources and close the block valves to maintain the pressure and retain the liquid.

• 10S to PE-101A and PE-101B (TV-2105 and TV-2106)

• 25S to PE-101C (TV-2107)

- Hot oil to PE-101E and F (TV-2101)

- b) Close the inlet valve of each user of LPW and MPW purge water.
- c) Close the discharge valves of PP-101A(B) and PP-102A, B(C).
- d) Lock the switch of each equipment that has been stopped.
- e) Confirm the level and pressure of each equipment in the control room and in the field.

2-2) Crystallizer

- a) Close the block valve of the following valve.

- FV-2302

- b) Close the inlet valves for HPW, MPW, and LPW purging and sealing waters.
- c) Close the discharge valve of PP-304A(B).
- d) Lock the switch of each equipment that has been stopped.
- e) Confirm the level and pressure of each equipment in the control room and in the field.

2-3) Separation and Drying

- a) Close the block valve of TV-2401.
- b) Close the inlet valves of LPW purging and sealing waters.
- c) Close the discharge valves of PP-402A(B) and PP-410.
- d) Lock the switch of each equipment that has been stopped.
- e) Confirm the level and pressure of each equipment in the control room and in the field.

2-4) Solvent and sealing and flushing

- a) Close the block valves of the following control valves to stop the heat source:

- TV-2502, TV-2503 and TV-2504.

- TV-2701, TV-2702, TV-2703 and TV-2706

- b) Close the inlet valve of each of the LPW, MPW, HPW

and DIW users.

c) Lock the switch of each equipment that has been stopped.

d) Confirm the level and pressure of each equipment in the control room and in the field.

2-5) Hot oil

a) Close the block valves of PF-601 main and pilot burners.

b) Lock the switches of PP-601A, B(C) and close the discharge valves.

2-6) Silo

Nothing special

2-7) PTA residue recovery

a) Close the inlet valves of PQ purging and sealing PQ.

b) Close the discharge valve of PP-902A(B).

c) Lock the switch of equipment that has been stopped.

d) Confirm the level and pressure of each equipment in the control room and in the field.

3) When power supply has been restored in a short period of time, the PTA Unit is started up in sequence, following the start-up procedure. The following are the major points to be performed in the start-up procedure.

Remarks: Refer to "4.1.2.1.10) on the actions to prevent of sedimentation of TA in the drum.

3-1) Confirmation of utilities

a) Receive DIW.

b) Establishment of the LPW, MPW and DIW systems

Remarks : Start-up of PP-502A(B), PP-503A(B) and PP-504A(B)

c) In the Drums, TA has deposited after the stop of the agitator. Therefore, start the agitator. In cases where the agitator are not started, add MPW or LPW available for dilution and blow some of the slurry

from the bottom of the Drum and try to start again.

Remarks: PD-101, PD-301 to PD-304, PD-401, and
PD-402, PD-901

d) Put the Hot oil system into circulation operation and
ignite PF-601.

e) Establish PE-101A to F.

Remarks: TIC-2101 287.0°C

f) Switch the suction of PP-102A, B(C) to LPW and start
PP-102A, B(C). (Parallel operation of two PP-102
units)

Remarks: FI-2201 109.8 t/hr

FIC-2201A approx. 54.9t/hr

FIC-2201B(C) approx. 54.9t/hr

4) In the case where the restoration of power supply is
expected to be delayed for a long time

4-1) Isolate each of PE-101A ~ F and PD-201.

4-2) Close the inlet valve of PD-201 and blow down the
holdups in PE-101A to F.

The holdup in PD-201 shall be drained to PD-301 by
opening PV-2201 slightly while keeping the pressure by
using H₂ compressor.

4-3) After the restoration of power supply, drain holdup in
the Crystallizer and downstream systems.

a) Make sure that the slurry and CTA conveying lines are
through.

4-4) Make sure that the Feed and Reactor system lines are
through by flushing them.

Remarks : PE-101A ~ F shall be washed with NaOH.

4-5) Establish hot water operation according to the start-up
procedure and flush the Feed, Reactor and Crystallizer
systems adequately. Then, resume CTA feed.

4.2.1.1.3 Emergency operations to be performed in the event of Electric Power Failure

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operations			
a)To maintain pressure and retain liquid.	a)After confirmation of the electric power failure, inform the operators of ESD. b)Operate the interlock switch of "PTA Unit All Shut Down". c)Close FV-2602 and FV-2601 on manual.	a)Return to the CCR. b)Close block valve of FV-2202.	
2)2nd operations			
a)To maintain pressure and retain liquid.	a)Start EDG system.	a)After start PP-502A(B) and PP-504A(B), start PJ-101, PJ-301~304, PJ-401, PJ-402 and PJ-901.	a)Close block valves of PF-601 main and pilot burners.
b)Turn off the heating and pressuring sources.	b)Operation to flow the FLW to the CCW Line. (This action is performed by operator of TA unit.)	b)Close the discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal.	b)Close discharge valve of each pump and lock the switches for the following stopped pumps and equipment. Also close block valve of mechanical seal.
c)Prevention of segimentation of each drum. Getting out the slurry in the Feed & Reactor system.	c)Switch off the rotary equipment having "MZ" on DCS. • PP-404A/B • PP-603 • PP-901 • TP-802A/B • PC-901 • PM-101-1,-2 • PM-901 • PM-902 • PM-405 • PZ-403	PJ-101, PJ-301, PJ-302, PJ-303 PJ-304, PP-502A/B, PP-504A/B PP-102A/B/C, PP-103A/B PP-503A/B, PP-101A/B, PC-201A/B PP-201A/B, PP-304A/B, PP-501A/B PM-101-1,2, PM-102, PM-105 PZ-100	PJ-401, PJ-901, PJ-402 PP-404-1, TP-802A/B, PJ-404 PC-601, PC-602, PX-401A/B/C-1,2,3 PP-601A/B/C, PP-402A/B PP-404A/B, PP-405A/B PP-406A/B, PP-410, PP-901 PM-401A/B/C, PM-402A/B PM-402PM(A/B), PM-404-1,-2, PM-901, PM-902, PZ-403 PZ-903, PC-402, PC-403A PC-901, PM-405, PP-902A/B PC-901OP
	d)Confirm the cutting off the heat source of each equipment.	c)Close block valves of following control valves: • TV-2101 • TV-2701, TV-2702, TV-2703 and TV-2706 • TV-2105, TV-2106, TV-2107 • TV-2502, TV-2503 and TV-2504 • FV-2302	c)Close block valve of following valve: • TV-2401
	e)Confirm the level and pressure of each equipment.	d)Start PP-103A(B) and then start PP-102A(B,C). After operate about 15 minutes, stop PP-102A(B) and PP-103A(B). e)Close inlet valve of each user to HPW, MPW, LPW and DIW purge waters.	d)Close inlet valve of each LPW user.

Remarks: Equipment which shall be started by EDG backup.

4.2.1.2 ESD of the^PTA unit

4.2.1.2.1 Summary

In case of the TA unit ESD, purified WG for pneumatic conveying system is stopped.

Therefore, operation of PTA unit must be switch to Recycle slurry operation.

Refer to Torouble shooting section 5.2.1.4.

4.2.1.2.2 Procedure

1) Switchover to recycle slurry operation.

Remarks: Carry out this operation according to the procedure of "2.2.5.3 Recycle Slurry Operation" in Various Work During Normal Operation.

2) Start the blower PC-402, after receive the N from OSBL. Normalize the operation of the dryer system.

Remarks: Refer to the procedure of "2.2.3.14 Switch the means of Pneumatic Conveying from PM-404 to PTK-800A, B and C from Purified WG to PC-402" in Various Work During Normal Operation.

3) Refer to the Table on the following pages.

Remarks: In case where the restoration of purified WG system is expected to be delayed for a long time, proceed to recycle solvent operation.

4.2.1.3.3 Emergency operations to be performed in the event of Electric Power Failure of 6.6 KV-TC-201 Feeders only

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operation			
a)Switchover to Recycle slurry operation.	a)Information the operators of Recycle Slurry Operation.	a)Return to the CCR.	
b)Keep the pressure of 7N.	b)Operate the sequence circuit of SC-2101 Normal to Recycle slurry changeover circuit.		
	c)Close FV-2405, FV-2411 and FV-2800.		
	d)Close KCV-2928. (7N valve of PM-901 bottom)		
2)2nd operation			
a)Restart of the dryer system for disposal of hold-up.	a)Receive the N from OSBL.		a)Start the blower and the dryer.
	b)Start the blower and dryer.		• Start PC-402.
	• Switch KCV-2804 to PC-402 side.		• Introduce the labyrinth N to the drayer.
	• Set PIC-2801A(B,C), PICS-2802 at 250mmH ₂ O each.		• Start PZ-403, PM-405 and PM-404-1,-2.
	• Start PC-402.		
	• Flow the N using on FV-2411.		
	• Switch the "RESET PZ-403" to on.		
	• Start PZ-403, PM-405 and PM-404-1,-2.		

4.2.2 Stop of 25S

4.2.2.1 Summary

After confirmation of operating condition, PTA Unit is shifted over Recycle Solvent Operation if required. Then, the PTA Unit is judged to shut down or restart.

4.2.2.2 Procedure

If the PTA Unit is shifted over shut down, its unit is performed Normal Shut Down Operation, when 25S is restored, its unit is performed from Recycle Solvent Operation to resume CTA feed.

4.2.3 Stop of 15S

4.2.3.1 Summary

In the event of the stop of supply of 15S, also 10S and 3S are stopped.

The main process system is brought into Recycle solvent operation.

4.2.3.2 Procedure

1)1st operation

1-1)Operate the Recycle solvent changeover switch.

Remarks: Continue the Recycle solvent operation for 15 minutes.

1-2)Operate the interlock switch of "PTA Unit All Shut Down."

1-3)Close block valve of FV-2202.

2)2nd operations

2-1)Operate the "Depower supply switch".

2-2)Close FCV-2602 on manual.

2-3)Feed and Reactor

a)Turn off the heating and pressuring sources and close the block valves to maintain the pressure and retain the liquid.

• 10S to PE-101A, B (TV-2105, TV-2106)

• Hot oil to PE-101E/F (TV-2101)

b)Close the inlet valve of each user of LPW and MPW purge waters.

c)Close the discharge valves of PP-101A(B) and PP-102A, B(C).

d)Lock the switch of each equipment that has been stopped.

e)Confirm the level and pressure of each equipment in the control room and in the field.

2-4)Crystallizer

a)Close the block valve of the following valve.

• FV-2302

- b)Close the inlet valves for HPW, MPW, and LPW purging and sealing waters.
- c)Close the discharge valve of PP-304A(B).
- d)Lock the switch of each equipment that has been stopped.
- e)Confirm the level and pressure of each equipment in the control room and in the field.

2-5)Separation and Drying

- a)Close the block valve of TV-2401.
- b)Close the inlet valves of LPW purging and sealing waters.
- c)Close the discharge valves of PP-402A(B) and PP-410.
- d)Lock the switch of each equipment that has been stopped.
- e)Confirm the level and pressure of each equipment on the control room and in the field.

2-6)Solvent and sealing and flushing

- a)Close the block valves of the following control valves to stop the heat source:
 - TV-2502, TV-2503 and TV-2504.
 - TV-2701, TV-2702, TV-2703 and TV-2706
- b)Close the inlet valve of each of the LPW, MPW, HPW and DIW users.
- c)Lock the switch of each equipment that has been stopped.
- d)Confirm the level and pressure of each equipment in the control room and in the field.

2-7)Hot oil

- a)Lock the switches of PP-601A, B(C) and close the discharge valves.

2-8)Silo

Nothing special.

- 3)When steam has been restored in a short period of time, the PTA Unit is started up in sequence. The following are the major points to be performed in the start-up procedure.

3-1)Confirmation of utilities

- a)Receive Steam. (15S, 10S, 3S, and 1S)

b) Establishment of the LPW, MPW and DIW systems.

Remarks : Start-up of PP-502A(B), PP-503A(B) and PP-504A(B)

c) In the Drums, TA has deposited after the stop of the agitator. Therefore, start the agitator.

Remarks: PD-101, PD-301 to PD-304, PD-401, PD-402 and PD-901

d) Put the hot oil system into circulation operation and ignite PF-601.

e) Establish PE-101A to E.

Remarks: TIC-2101 287.0°C

f) Switch the suction of PP-102A, B(C) to LPW and start PP-102A, B(C). (Parallel operation of two PP-102 units)

Remarks: FI-2201 109.8 t/hr

FIC-2201A 54.9 t/hr

FIC-2201B(C) 54.9 t/hr

4) In case where the restoration of steam is expected to be delayed for a long time

4-1) Isolate each of PE-101A ~ F and PD-201.

4-2) Blow down holdup in each block valve to the sewer.

4-3) After the restoration of steam, drain holdup in the Crystallizer and downstream systems.

a) Make sure that the slurry and CTA conveying lines are through.

4-4) Make sure that the Feed and Reactor system lines are through by flushing them.

Remarks: PE-101A ~ F shall be washed with NaOH.

4-5) Establish hot water operation according to the start-up procedure and flush the Feed, Reactor and Crystallizer systems adequately. Then, resume CTA feed.

4.2.3.3 Emergency operations to be performed in the event of stop of 15S

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operations			
a)Switch to Recycle solvent operation.	a)Operate the Recycle solvent changeover switch.	a)Return to the CCR.	
b)After 15 minutes, to maintain pressure and retain liquid.	b)After operation of Recycle Solvent, in succession,inform the operators of ESD.	b)Close block valve of FV-2202.	
	c)Operate the interlock switch of "PTA Unit All Shut Down".		
	d)Operate the "Depower supply switch".		
2)2nd operations			
a)Stop rotary equipment	a)Close FCV-2602 on manual.	a)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal.	a)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal.
b)Turn off the heating and pressuring sources.	b)Confirm the level and pressure of each equipment in the control room.	PP-101A/B, PP-102A/B/C PC-201A/B, PP-201A/B PP-304A/B, PP-501A/B PM-101-1,-2, PM-102 PP-103A/B, PM-105 PZ-100	PP-402A/B, PP-404A/B, PP-405A/B PP-406A/B, PP-410, PP-901 PM-401A/B/C, PM-402A/B PM-404-1,-2, PM-901, PM-902 PM-405, PZ-403, PC-402 PC-403A, PC-901, PP-902A/B
		b)Close block valves of following control valves:	b)Close block valve of following valve:
		• TV-2101 • TV-2105, TV-2106 and TV-2107 • TV-2701, TV-2702, TV-2703 and TV-2706 • TV-2502, TV-2503 and TV-2504 • FV-2302	• TV-2401
		c)Close inlet valve of each user to HPW, MPW, LPW and DIW purge waters.	c)Close inlet valve of each LPW user.

4.2.4 Stop of IA

4.2.4.1 Summary

In the event of the stop of supply of IA, the control valves move to such position that safety is ensured, and the process systems get into the state in which the pressure is maintained and the liquid is retained.

4.2.4.2 Procedure

1) 1st operation

- 1-1) Operate the interlock switch of "PTA Unit All Shut Down".

Remarks: Refer to the interlock circuit for the detailed working conditions.

- 1-2) Operate the "Depower supply switch".

- 1-3) Close FV-2602 on manual.

- 1-4) Close the block valve of FV-2202 in the H₂ feed line.

2) 2nd operation

- 2-1) Refer to the Table on the following Pages.

- 3) When the supply of IA is restored in a short period of time.

3-1) Receive utilities.

- a) Receiving of 3S, 10S and 25S and establishment of the operation of Flash Drum system.

- b) Receiving of N.

- 3-2) Establishment of the LPW, MPW, HPW and DIW systems.

- a) Start PP-502, PP-503 and PP-504.

- b) Introduce purge water to each user.

- 3-3) Make sure that there is no abnormality in PJ-101, PJ-301, PJ-302, PJ-303, PJ-304, PJ-401, PJ-402, PJ-404 and PJ-901.

- 3-4) Make sure that there is no abnormality in Feed & Reactor System.

- 3-5) After the circulation operation of the hot oil system,

ignite PF-601 to heat up hot oil.

3-6) Start up the Centrifuges, RVF and the Dryer.

a) The discharge of PM-404 shall be switched to the Run Down Silo TTK-401.

Remarks: The Centrifuges and feed lines shall be purged thoroughly.

3-7) Introduce 10S into PE-101A & B. 25S shall be introduced into PE-101C.

Also, start heating up PE-101E & F by introducing Hot oil into them.

3-8) Switch the suction of PP-102 to LPW and start PP-102.

After performing the above operation, the process system is brought back into operation, following the start-up procedure. CTA feed is resumed after performing an adequate flushing of the Feed, Reactor and Crystallizer systems and confirming the action of the control valves.

4) In cases where the restoration of supply of IA is delayed

4-1) Blow down the holdup in the Feed and Reactor system to the outside of the process system. The holdup in PD-201 shall be drained to PD-301 by opening PV-2201 slightly while keeping the pressure by using H₂ compressor.

4-2) Introduce water into the mechanical seals of PJ-301, PJ-302, PJ-303, PJ-304, and PJ-401.

a) Start PP-504, and close the block valve of PV-2504 and then control the pressure of DIW system by the by-pass valves of PV-2504.

b) Introduce DIW to each agitator and control the pressure by use of the by-pass line at the outlet side of agitator.

4-3) In cases where the restoration of supply of IA is expected to be delayed for a further longer time, blow down holdup from all the systems beginning from the Crystallizers.

4.2.4.3 Emergency operations to be performed in the event of stop of IA

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operations			
a)To maintain pressure and retain liquid.	a)Information the operators of stoppage of IA. b)Operate the interlock switch of "PTA Unit All Shut Down". c)Operate the "Depower supply switch".	a)Return to the CCR. b)Close block valve of FV-2202.	
2)2nd operations			
a)To maintain pressure and retain liquid.	a) Close FCV-2602 on manual.	a)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal. PP-101A/B PP-102A/B/C PC-201A/B, PP-201A/B PP-304A/B PP-501A/B PM-101-1, 2, PM-102 PM-105, PZ-100, PP-103A/B	a)Close block valves of PF-601 main and pilot burners. b)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal. PP-402A/B, PP-404A/B, PP-405A/B PP-406A/B PP-410, PP-901 PM-401A/B/C, PM-402A/B PM-404-1,-2, PM-901, PM-902 PZ-403, PC-402, PC-403A PC-901, PP-902A/B, PM-405
b)Turn off the heating and pressuring sources.	b)Confirm the level and pressure of each equipment in the control room.	b)Close block valves of following control valves: · TV-2101 · TV-2701, TV-2702, TV-2703 and TV-2706 · TV-2105, TV-2106 and TV-2107 · TV-2502, TV-2503 and TV-2504 · FV-2302 c)Close inlet valve of each user to HPW, MPW, LPW and DIW purge waters.	c)Close block valve of following valve: · TV-2401 d)Close inlet valve of each LPW user.

4.2.5 Stop of 9N

4.2.5.1 Summary

N is used for keep the pressure of PM-401, PD-401 and PD-403 system by PV-2401A.

In the event of stop of N, operation is continued because it is possible to introduce 7N to drain valve of PG-2407 (PD-403 top) by connecting the hose temporarily.

4.2.5.1 Procedure

1) Stopping receiving 9N

1-1) Confirm the pressure drop of 9N receiving line.

1-2) To introduce purified WG to drain valve of PG-2407 (PD-403 top) by connecting the hose temporarily.

1-3) For another equipment which is using N, short period of time does not raise any problem, continue normal operation.

4.2.6 Stop of CCW

4.2.6.1 Summary

In cases where the supply of CCW stops, the emergency shut down operation is done as follows.

4.2.6.2 Procedure

1) 1st Operations

- 1-1) Operate the interlock switch of "PTA Unit All Shut Down".
- 1-2) Operate the "Depower supply switch".
- 1-3) Close FV-2602 on manual.
- 1-4) Stop PP-601A, B, (C).
- 1-5) Close the block valve of FV-2202 in the H₂ feed line.
- 1-6) Introduce FLW to CCW of ESD line for agitators.
 - a) Close HZV-3301.
 - b) Open HZV-3302.

Remarks: It is possible to operate each agitator and pumps.

2) 2nd operation

- 2-1) Refer to the Table on the following pages.
- 3) In cases where the operation of the CCW system is restored in a short period of time
 - 3-1) Make sure that CCW has been introduced into each user.
 - 3-2) Start the circulation operation of the Hot oil system to heat it up.
 - a) Ignite PF-601 and raise the temperature of the Hot oil system.
 - 3-3) Start the Dryer system.
 - a) If there is not a sufficient supply of purified WG, start PC-402.
 - 3-4) Start PM-401 and PM-402.
 - a) After disposing the holdup from PD-401, PD-402, Crystallizers and Drums, restart PM-401 and PM-402,

following the start-up procedure.

3-5) Flush the Feed and Reactor system sufficiently by
conducting LPW operation.

4.2.6.3 Emergency operations to be performed in the event of stop of CCW

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operations			
a)To maintain pressure and retain liquid.	a)Inform the operators of Shut Down of the PTA unit. b)Operate the interlock switch of "PTA Unit All Shut Down". c)Operate the "Depower supply switch". d)Close FV-2602 on manual. e)Intorduce FLW to CCW of ESD Line. · Close HZV-3301. · Open HZV-3302.	a)Return to the CCR. b)Close block valve of FV-2202.	a)Stop PP-601A,B(C). (Close HCV-2602A, B(C))
2)2nd operations			
a)To maintain pressure and retain liquid. b)Turn off the heating and pressuring sources.	a)Confirm the level and pressure of each equipment in the control room. b)Start PJ-101, PJ-301~304, PJ-401 and PJ-402. (After establishcment of CCW ESD line)	a)Stop the following agitators and pumps. · PJ-101, PJ-301, PJ-302, PJ-303, PJ-304, PJ-401, PJ-402, · PP-502A,B, PP-503A,B, PP-504A,B (Close discharge valve) b)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal. · PP-101A/B · PP-102A/B/C · PC-201A/B, · PP-201A/B · PP-304A/B · PP-501A/B · PM-101-1, 2, · PM-102, · PM-105 · PZ-100 · PP-103A/B c)Close block valves of following valves: · TV-2101 · TV-2701, TV-2702, TV-2703 and TV-2706 · TV-2105, TV-2106 and TV-2107 · TV-2502, TV-2503 and TV-2504 · FV-2302 d)Close inlet valve of each user to HPW, MPW, LPW and DIW purge waters.	a)Close block valves of PF-601 main and pilot burners. b)Close discharge valve of each pump and lock the switches for the following pumps and equipment. Also close block valves of mechanical seal. · PP-402A/B · PP-404A/B · PP-405A/B · PP-406A/B · PP-410 · PP-901 · PM-401A/B/C · PM-402A/B · PM-404-1,-2 · PM-405 · PM-901 · PM-902 · PZ-403 · PC-402 · PC-403A · PC-901 · PP-902A/B c)Close block valve of following valve: · TV-2401 d)Close inlet valve of each LPW user.

4.2.7 Stop of FLW

4.2.7.1 Summary

In the event of stop of FLW, the operation of pumps and seal pots for a short period of time does not raise any problem, continuing normal operation.

In the event of start TP-802 before introduce DIW for flashing the mechanical seal of TP-802 by connecting the hose.

Remarks: In the event of start TP-800 refer to CTA ESD MANUAL.

1) Stopping receiving FLW

1-1) Confirm the pressure drop of FLW receiving line.

2) Changeover from FLW to DIW

2-1) Connect the DIW hose to the mechanical seal of TP-802.

4.2.8 Stop of DIW

4.2.8.1 Summary

Since DIW is being used directly in the process as the solvent for the PTA Unit, it is impossible to continue operation for a long period of time after stop of DIW. In the event of stop of DIW, therefore, CTA feed is quickly stopped, and the Feed and Reactor system is brought into the state of maintaining the pressure and retaining liquid after being flushed.

4.2.8.2 Procedure

Immediately stop CTA feed and flush the process by carrying out recycle solvent operation. When there is no more solvent in PD-502 and PD-501, conduct an emergency shut-down.

1) Procedure

Refer to the Table on the following pages.

2) When the supply of DIW is restored, DIW shall be introduced into each user and each system is brought back into operation after being flushed with it.

However, all the slurry lines of the Feed, Reactor and Crystallizer systems, PT-401, PD-402, etc., shall be checked to be sure that they are through by using purge water, etc. because there is the possibility that they are plugged due to the deposition of CTA.

3-1) Check each of the DIW, LPW, MPW and HPW lines and then introduce each flushing water into it to make sure that it is through.

a) Charge PD-502 with DIW.

- Set the degree of opening of LIC-2502 on manual and receive DIW. When the normal level is reached, place LIC-2502 on automatic control.

b) Confirm that LPW has been introduced.

- Start PP-502 and set PIC-2502 at 12kg/cm²G.

- Introduce water into each LPW user.
 - c) Charge PD-501 with LPW.
 - Open LIC-2501B on manual and charge LPW to PD-501.
 - d) Confirm that MPW has been introduced.
 - Start PP-503 and set PIC-2503 at 53kg/cm²G.
 - Introduce water into each MPW user.
 - e) Confirm that mechanical seal lines of agitators have been line-up.
 - Start PP-504 and set PIC-2504 at 16 kg/cm²G.
 - Introduce DIW to mechanical seal of each agitators.
- 3-2) After confirming that the Dryer system is in operation or starting it, start the feed to PM-402 and begin processing the slurry.
- a) Bring the PT-401 system into normal conditions.
 - Set FIC-2408 at 14.0t/hr.
 - b) Start N to the Dryer system.
 - Set FIC-2411 at 1,500Nm³/hr.
- Remarks: Check PI-2411 to be sure that the PT-401 tray is not clogged.
- c) Establish a flow of 2,200 Nm³/hr on FICS-2405.
 - d) Start PM-404, Screw Feeder of PM-404, PM-405 and PZ-403.
 - e) Start PM-402.
 - Confirm that PM-402 ~ PD-405 system is in operation.
 - Open FV-2402 on manual and perform LPW purging from the suction side of PP-402 for 2 to 3 minutes. After purging, close them.
 - Start PP-402.
 - Start feed by opening FV-2402 on manual and dispose of the holdup.
- 3-3) Start PM-401 and dispose of the holdup in the Crystallizers.
- Start PM-401.

- Introduce purge water into the feed tube.
- Introduce reslurry water.
- Open LV-2304 on manual and perform LPW purging from the suction side of PP-304 for 2 to 3 minutes. After purging, close them.
- Start PP-304.
- Start feed by opening LV-2304 on manual and dispose of the holdup.

3-4) Start the operation of PD-403 and PD-404.

3-5) Ignite the PF-601 main burners and heat up hot oil.

3-6) Heat of PE-101A to F.

3-7) Perform the LPW flushing of PE-101A to F~ PP-102 ~ PD-201 ~ PD-301.

- Open HZV-2106 for charging LPW which is on the inlet side of PE-101A.

3-8) After finishing carrying out the step 3-7) above, bring the system back into operation, following the start-up procedure in sequence.

4.2.8.3 Emergency operations to be performed in the event of stop of DIW

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st operatios			
a)Switch to recycle solvent operation.	a)Operate the Recycle solvent Changeover switch.	a)Return to the CCR.	
b)Hold the slurry to PD-301 ~ PD-304 and etc.	b)After operation of Recycle Solvent, in succession, inform the operators of stoppage of DIW.	b)Close inlet, outlet block valves of mechanical seal water. · PJ-301, PJ-302, PJ-303, PJ-304, PJ-401	
c)After 10 minutes switch to All SD.	c)Close FV-2401A, B(C) and close LV-2304A, B(C).	c)Close block valve of FV-2202.	
d)To maintain pressure and retain liquid.	d)Operate the interlock switch of "PTA Unit All Shut Down".		
	e)Operate the "Depower supply switch".		
	f)Close FCV-2602 on manual.		
2)2nd Operations			
a)To maintain pressure and retain liquid.	a)Confirm the level and pressure of each equipment in the control room.	a)Close discharge valve of each pump and lock the switches for the following stopped pumps and equipment. Also close block valve of mechanical seal. · PP-101A/B · PP-102A/B/C · PC-201A/B · PP-304A/B · PP-501A/B · PM-101-1, 2 · PM-102 · PM-105 · PZ-100 · PP-201A/B · PP-103A/B	a)Close block valves of PF-601 main and pilot burners.
b)Turn off the heating and pressuring sources.		b)Close block valves of following control valves: · TV-2101 · TV-2701, TV-2702, TV-2703 and TV-2706 · TV-2105, TV-2106 and TV-2107 · TV-2502, TV-2503 and TV-2504 · FV-2302 c)Close inlet valve of each user to HPW, MPW, LPW and DIW purge waters.	b)Close discharge valve of each pump and lock the switch for the following stopped pumps and equipment. Also close block valves of mechanical seal. PP-402A/B, PP-404A/B, PP-405A/B PP-406A/B, PP-410, PP-901 PM-401A/B/C, PM-402A/B PM-404-1,2, PM-901, PM-902 PZ-403, PC-402, PC-403A PC-901, PP-902A/B, PM-405 c)Close block valve of following valve: · TV-2401 d)Close inlet valve of each LPW user.

4.2.9 H₂ Failure

4.2.9.1 Summary

When the supply of H₂ to PD-201 stops, the Hydrogenation Reaction can not be carried out in the Reactor, then, resulting in producing off-spec product.

Therefore, CTA feed is stopped and Recycle Solvent operation is continued waiting the restoration of H₂ supply.

4.2.9.2 Procedure

1) 1st Operation

1-1) Stop CTA feed and proceed to recycle solvent operation.

1-2) Switch H₂ to N at the suction of PD-202.

2) 2nd Operation

2-1) After 1 hour from stopping CTA feed, switch the destination of the discharge of PM-404 from PTK-800A, B or C to TTK-401.

Remarks: Above procedures are carried out according to the normal shutdown procedure.

4.2.9.3 Emergency operations to be performed in the event of stop of H₂

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st Operations			
a)Switch to Recycle Solvent operation	a)Operate the Recycle Solvent Changeover Switch. b)Inform the operator of stoppage of H ₂ .	a)Return to the CCR. b)Switch H ₂ to N at the suction of PD-202. (SP-2213(N) ; Change to open side)	
2)2nd Operations			
a)Switch to Run Down silo. (After 1hour)	a)Operate the SC-2800 PTA silo changeover circuit. (PTK-800A/B/C → TTK-401)		

4.2.10 Hot Oil Failure

4.2.10.1 Summary

When the supply of Hot oil to each Hot oil user stops, the dissolution of feed slurry becomes inadequate and CTA begins to deposit, causing the plugging of the lines and equipment and the malfunction of the control valves. In such event, CTA feed is immediately stopped and operation is switched to recycle solvent operation to flush the process system. In cases where the trouble is very serious to continue operation, shut down the process using "PTA Unit All Shut down" interlock and "Depower supply switch".

4.2.10.2 Procedure

1) The following are the conceivable causes of Hot oil failure:

- a) Trouble with the PF-601 fuel system.
- b) Trouble with the Hot oil circulation system.
- c) Instrument, electrical or mechanical trouble with the Hot oil system.

2) Procedure

Refer to the Table on the following pages.

4.2.9.3 Emergency operations to be performed in the event of stop of Hot Oil

General Operations	Board-man	Field Operator in Charge of Feed, Reactor and Solvent System	Field Operator in Charge of Separation, Dryinbg and Hot Oil System
1)1st Operations			
a)Switch to Recycle Solvent operation	a)Operate the Recycle Solvent Changeover Switch. b)Inform the operator of stoppage of Hot oil.	a)Return to the CCR. b)Switch H ₂ to N at the suction of PD-202. (SP-2213(N) ; Change to open side)	
2)2nd Operations			a) Prepare for the restart of PF-601.
a)Switch to Run Down silo. (If PTA is off-spec by checking the quality.)	a)Operate the SC-2800 PTA silo changeover circuit. (PTK-800A/B/C → TTK-401)		